
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

20 Aug, 2026

Tutorial-3

3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
 2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
 3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
 4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
 5. Submissions will not be accepted after the tutorial hours.
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Q1. Negative Binomial Distribution Consider a sequence of independent Bernoulli trials, where each trial results in either a success with probability p or a failure with probability $(1 - p)$. Let X denote the number of trials needed to obtain r successes.

- (a) Determine the probability mass function of X .

Hint: What must happen in first $x - 1$ trials and what must happen on the x -th trial.

$$p_X(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r, \quad x = r, r+1, r+2, \dots$$

- (b) Show that when $r = 1$, Negative Binomial reduces to the Geometric distribution.

Hint: Write the resulting PMF and identify what X represents in this special case.

Q2. Hypergeometric Distribution A population contains N objects, of which exactly K are classified as favorable and $N - K$ are classified as unfavorable. A sample of n objects is selected **without replacement**. Let X denote the number of favorable objects obtained in the sample.

- (a) Determine the range of X .
(b) Derive the probability mass function of X .

Hint: Count the number of ways to select x favorable objects and $n - x$ unfavorable objects from the population.

$$x = \max\{0, n + K - N\}, \dots, \min\{K, n\}.$$
$$p_X(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = \max\{0, n + K - N\}, \dots, \min\{K, n\}.$$

- (c) Explain why the Hypergeometric distribution is generally **not** a Binomial distribution. In particular, identify which assumption of the Binomial model is violated when sampling is performed without replacement.

Q3. Mean and Variance of Bernoulli Trials

A Bernoulli trial has two possible outcomes: Success (with probability p) and Failure (with probability $1 - p$). Let the random variable X be defined as

$$X = \begin{cases} 1, & \text{if the trial results in Success,} \\ 0, & \text{if the trial results in Failure.} \end{cases}$$

- (a) Write down the probability mass function of X .
- (b) Derive the mean, $\mathbb{E}[X]$, from the definition of expectation.
- (c) Derive the variance, $\text{Var}(X)$, by first calculating $\mathbb{E}[X^2]$ and then using

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Q4. Mean and Variance of Binomial Distribution

A manufacturing process produces independent components, each of which is defective with probability p . A quality engineer inspects n components. Let X denote the number of defective components.

- (a) State the probability mass function of X .
- (b) Using the identities

$$\mathbb{E}[X] = \sum_{x=0}^n x p_X(x)$$

and

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

derive the mean and variance of the Binomial distribution.

- (c) Explain how the mean and variance change as the success probability p increases from 0 to 1 while keeping n fixed.

Q5. Mean and Variance of Geometric Distribution

An engineer repeatedly tests identical electronic components until the first defective component is observed. Each component is defective independently with probability p . Let X denote the trial on which the first defective component is detected.

- (a) Write down the probability mass function of X .
 - (b) Using the definitions of expectation and variance, derive the mean and variance of the Geometric distribution.
 - (c) Explain why the expected value increases as the probability of success p decreases.
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